



Contact

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Location: Lyon, France
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Portfolio: Portfolio

Soft Skills

- Teamwork
- Working autonomy
- Technical communication
- Adaptability to new tools
- Proactive

Technical Skills

CFD: Ansys Fluent
CSM: Patran/Nastran
CAD: CATIA
Programming: Python,
Matlab, C/C++, HTML
HPC: parallel computing
Project tools: Git/GitHub

Languages

French: Native
English: C1 – TOEIC 945

Interests

- Football (playing and coaching)
- Badminton
- Travel

Romain Urbaniak

Master's student in Mechanical Engineering at the École Polytechnique Universitaire de Lyon

Seeking a final internship in fluid mechanics applied to aeronautics
Available from March to August 2026 (22 weeks)

Professional Experience

● *Engineering Intern, Japan* 2025

TOHOKU University, Sendai, Japan
Aeroelastic analysis, reduced-order modeling, and optimization of composite aircraft wing geometries.

Technical skills gained:

Python, CFD/CSM, data analysis, reduced-order modeling.

Transversal skills developed:

- Planning and prioritizing simulations to meet project milestones
- Adapting to an international environment
- Collaborating with researchers from different disciplines
- Presenting results in academic meetings
- Writing scientific reports in English

Valorization :

- **Author:** “Data-Driven Model-Order Reduction of Optimized Wing Geometries of Composite Aircraft”, R. Urbaniak, Y. Abe, ICFD 2025 Conference, Sendai, Japan
- **Co-author:** “Design Optimization of Composite Aircraft Wings Considering Coupled Aeroelasticity and Geometrical Nonlinearity”, Y. Liu, Y. Yatsu, R. Urbaniak, Y. Abe, CM3 – Transport 2025, Trieste, Italy, 2025

● *Event Staff* 2023–Present

● *Waiter / Bartender* Summer 2023/2024

Education

● *Mechanical Engineering Master* 2023–Present

École Polytechnique Universitaire de Lyon, Lyon, France

Specialization:

Environmental and Industrial Fluid Mechanics,
High-Performance Numerical Simulation,
Turbomachinery.

- Vibration study of a helicopter blade (Matlab)
- Heat equation resolution (Matlab)
- Aircraft wing study (C programming)
- Bicycle luggage rack design (CATIA)

● *Preparatory Class (PSI)* 2020–2023

Lycée St Joseph Pierre Rouge, Montpellier, France

- Parking sensor: precision improvement (Python)
- Mars air replenishment (Python)

● *High School Diploma – Honours* 2019

Lycée Théodore Aubanel, Avignon